Indiana University Indianapolis
MATH-I 241 Practice Test 3
Derivatives

Prof. Ramras

Name: _____

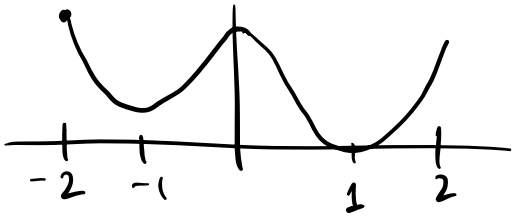
Instructions

- Test 3 will cover Sections 3.1, 3.2, 3.3, 3.4, 3.5, and 3.7. Material from 3.7 will be similar to quiz problems - that is, very close to the assigned HW for 3.7.
- No cell phones, calculators, watches, technology, or hats.
- Print your name on this cover page.
- This test is closed book and closed notes.
- You may not go to the bathroom during the test.
- Show all work for partial credit.
- If you wish for something not to be graded, strike it out neatly.
- Clearly indicate your final answer.

No calculators, notes, or books.

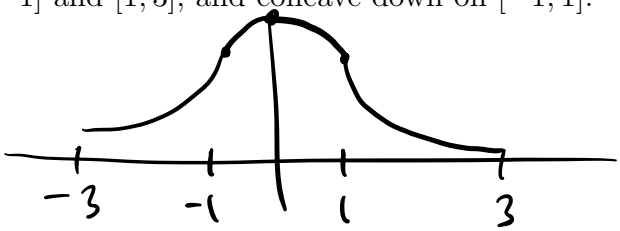
1. Sketch the graph of a function that is continuous on $[-2, 2]$, has an absolute maximum at -2, a local minimum at -1, a local maximum at 0, and an absolute minimum at 1.

One possible answer:



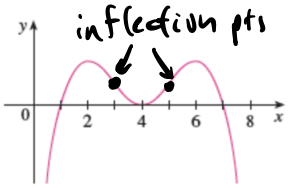
2. Sketch the graph of a function that is continuous on $[-3, 3]$ and is increasing on $[-3, 0]$, decreasing on $[0, 3]$, concave up on $[-3, -1]$ and $[1, 3]$, and concave down on $[-1, 1]$.

One possible answer:



3. The graph of a function $y = f(x)$ is shown below. Use this graph to determine the intervals on which $y = f(x)$ is increasing, the intervals on which $y = f(x)$ is decreasing, the x–coordinates of the local maxima and local minima of $y = f(x)$, the intervals on which $y = f(x)$ is concave up, the intervals on which $y = f(x)$ is concave down, and the x–coordinates of the inflection points of $y = f(x)$.

Note: I'll assume domain = (0,8)



Increasing on $(0,2]$, $[4,6]$ loc. max @ $x = 2, 6$

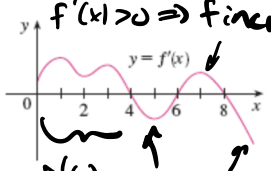
Decreasing on $[2,4]$, $[6,8)$ loc. min @ $x = 4$

Concave up on $[1,5]$ infl. pts @ $x = 1, 3$

Concave down on $[0,1]$, $[5,8)$

Note: it's fine to write these intervals with or w/o endpts

4. The graph of $y = f'(x)$ (the derivative of a function $y = f(x)$) is shown below. Use this graph to determine the intervals on which $y = f(x)$ is increasing, the intervals on which $y = f(x)$ is decreasing, the x–coordinates of the local maxima and local minima of $y = f(x)$, the intervals on which $y = f(x)$ is concave up, the intervals on which $y = f(x)$ is concave down, and the x–coordinates of the inflection points of $y = f(x)$.



$f'(x) > 0 \Rightarrow f$ incr. on $[6,8]$

$f'(x) > 0$ ⇓ f incr. on $[0,4]$

$f'(x) < 0$ ⇓ f decr. on $[4,6]$ & $[8,\infty)$

Loc. maxima: places where f changes from incr. to decreasing: $x = 4, 8$

Loc. minima: where f' changes decr → incr.: $x = 6$

Concave up: where f' is increasing: $[0,1]$, $[2,3]$, $(5,7]$

Concave down: where f' is decreasing: $[1,2]$, $[3,5]$, $[7,\infty)$

infl. pts: $x = 1, 2, 3, 5, 7$

5. Let $f(x) = 2x^3 - 9x^2 + 12x - 3$.

(a) Find the critical numbers of f.

$f'(x) = 6x^2 - 18x + 12 = 0$

$6(x^2 - 3x + 2) = 0$

$6(x-2)(x-1) = 0 \rightsquigarrow x = 1, 2$

(b) For each critical number you found in part a), determine whether it is a local maximum of f, a local minimum of f, or neither.

Sign chart for f': $+$ on $(-\infty, 1)$, $-$ on $(1, 2)$, $+$ on $(2, \infty)$

$\Rightarrow$ $x=1$ is a loc. max, $x=2$ is a loc min

Or: $f''(x) = 12x - 18$ so $f''(1) = -6 < 0$

$\Rightarrow$ loc max @ 1

$f''(2) = 6 > 0$

$\Rightarrow$ loc min @ 2

(c) Find the absolute maximum and absolute minimum values of $f(x)$ on the interval $[0, 3]$.

endpts: $f(0) = -3$, $f(3) = 2 \cdot 3^3 - 9 \cdot 3^2 + 12 \cdot 3 - 3$

$= 2 \cdot 27 - 3 \cdot 27 + 36 - 3$

$= -27 + 33 = 6$

Crit pts: $f(1) = 2 - 9 + 12 - 3 = 2$

$f(2) = 2 \cdot 2^3 - 9 \cdot 2^2 + 12 \cdot 2 - 3 = 16 - 36 + 24 - 3$

$= -20 + 21 = 1$

Ab max: 6, Ab min -3

(d) Find the intervals on which f is increasing and the intervals on which f is decreasing.

incr: $(-\infty, 1)$, $(2, \infty)$

decr: $(1, 2)$

(e) Find the intervals on which f is concave up and the intervals on which f is concave down, and find the inflection points of f.

$f' = 12x - 18 = 6(2x-3)$ $f''(x) = 0$ when $x = 3/2$

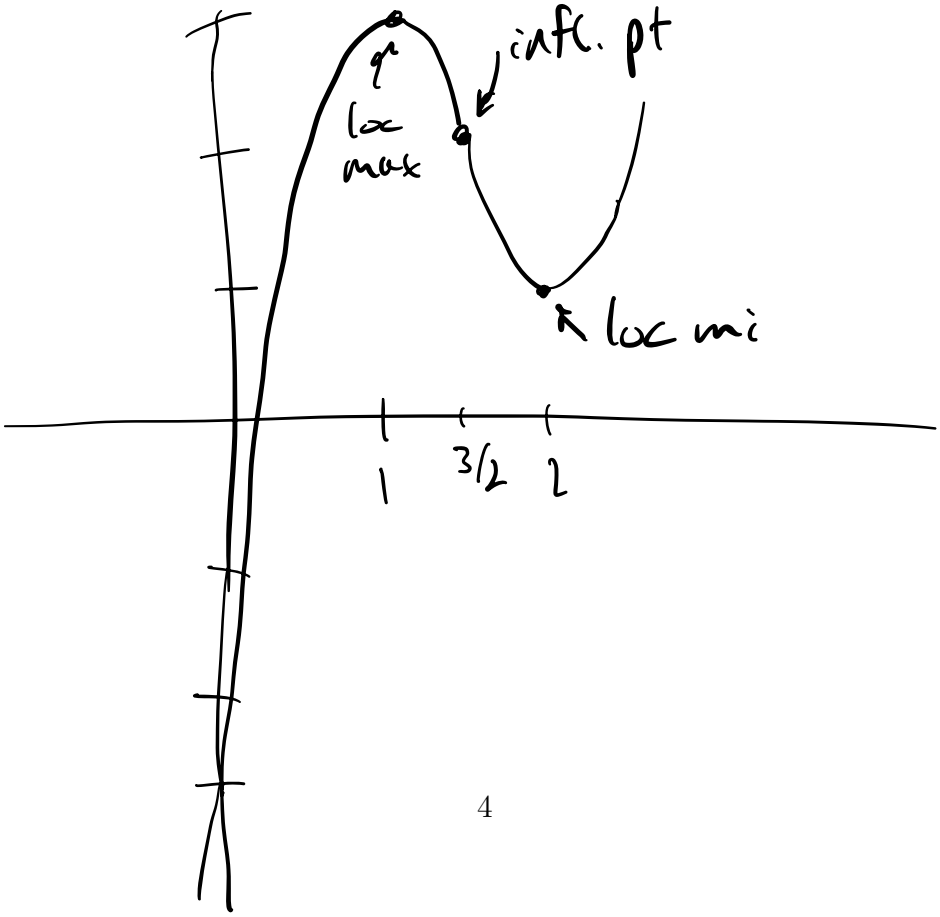
Sign chart: $-$ | $+$ with $3/2$

concave down on $(-\infty, 3/2)$

Concave up on $(3/2, \infty)$

inflection pt: $x = 3/2$

(f) Sketch the graph of $y = f(x)$, showing all the features above. Make sure to label the local maxima, local minima, and inflection points.



6. Find the absolute maximum and absolute minimum values of $f(x) = \frac{x}{x^2-x+1}$ on the interval $[0, 3]$.

$f'(x) = \frac{1(x^2-x+1) - x(2x-1)}{(x^2-x+1)^2} = 0 \iff x^2-x+1 - x(2x-1) = 0$

$\iff -x^2+1 = 0 \iff 1 = x^2 \iff x = \pm 1$

Note: $x = -1$ is not in $[0,3]$

Endpts: $f(0) = 0$, $f(3) = \frac{3}{9-3+1} = \frac{3}{7}$

Crit pts in $[0,3]$: $f(1) = \frac{1}{1-1+1} = 1$

Ab max: 1 @ $x = 1$

Ab min: 0 @ $x = 0$

7. Explain why the function $y = \sin^2(x)$ satisfies the hypotheses of Rolle's Theorem on the interval $[0, \pi]$. Then find all numbers that satisfy the conclusion of Rolle's Theorem.

$\sin^2(x)$ is differentiable & cont. because it is a composition of the diff'ble fcns $y = \sin(x)$ & $y = x^2$, and $\sin^2(0) = 0 = \sin^2(\pi)$. Rolle's then says $y'(x) = 0$ for some $x \in (0, \pi)$. $y'(x) = 2\sin(x)\cos(x)$, so $y'(x) = 0$ when $\sin(x) = 0$ & when $\cos(x) = 0$. So $x = \pi/2 \in (0, \pi)$ is the only solution.

8. Explain why the function $y = 1/x$ satisfies the hypotheses of the Mean Value Theorem on the interval $[1, 3]$. Then find all numbers that satisfy the conclusion of the Mean Value Theorem.

$y = 1/x$ is a ration fcn w/ $[1,3]$ in its domain so it is cont. & diff'ble.

$\frac{y(3)-y(1)}{3-1} = \frac{1/3 - 1}{2} = \frac{-2/3}{2} = -1/3$

$y'(x) = -x^{-2}$, so need to solve $-x^{-2} = -1/3$, i.e. $\frac{1}{x^2} = 1/3$, $x^2 = 3$, $x = \pm\sqrt{3}$. The only soln in $(1,3)$ is $\sqrt{3} \approx 1.732\ldots$

9. Let $f(x) = \dfrac{x}{x^2+1} + 1$.

(a) Find the intervals on which f is increasing and the intervals on which f is decreasing.

$$f'(x) = \frac{1\cdot(x^2+1) - x(2x)}{(x^2+1)^2} = 0 \iff x^2+1-2x^2 = 0$$
$$\iff 1-x^2 = 0$$
$$\iff x = \pm 1$$

Sign chart: $-$ | $+$ | $-$ (at -1, 1)

f is incr. on $[-1,1]$ & decr. on $(-\infty,-1)$, $(1,\infty)$

(b) Find local maximum and minimum values of f.

f has a local max at $x=1$, and $f(1) = \frac{1}{1+1}+1 = 3/2$

f " " " min " $x=-1$, and

$f(-1) = \frac{-1}{1+1}+1 = 1/2$.

(c) Find the intervals on which f is concave up and the intervals on which f is concave down, and find the inflection points of f.

$$f''(x) = \frac{d}{dx}\,\frac{1-x^2}{(x^2+1)^2} = \frac{-2x(x^2+1)^2 - (1-x^2)\cdot 2(x^2+1)\cdot 2x}{(x^2+1)^4}$$

$f''(x) = 0$ when $-2x(x^2+1)^2 - 4(1-x^2)(x^2+1)\cdot x = 0$.

Factoring out $-2x(x^2+1)$, which is only 0 when $x=0$,

we get $(x^2+1) + 2(1-x^2) = -x^2+3 = (\sqrt{3}+x)(\sqrt{3}-x)$

So $f''(x)$ changes sign @ $x=-\sqrt{3}$, $x=0$, & $x=\sqrt{3}$:

$-2x(x^2+1) > 0$ ↗ $\sqrt{3}+x<0$, $\sqrt{3}-x>0$

$-$ | $+$ | $-$ | $+$ (at $-\sqrt{3}$, 0, $\sqrt{3}$)

CD: $(-\infty,-\sqrt{3})$, $(0,\sqrt{3})$

CU: $(-\sqrt{3},0)$, $(\sqrt{3},\infty)$

infl pts: $x = -\sqrt{3}, 0, \sqrt{3}$

(d) Find $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$ and determine the horizontal asymptotes of f.

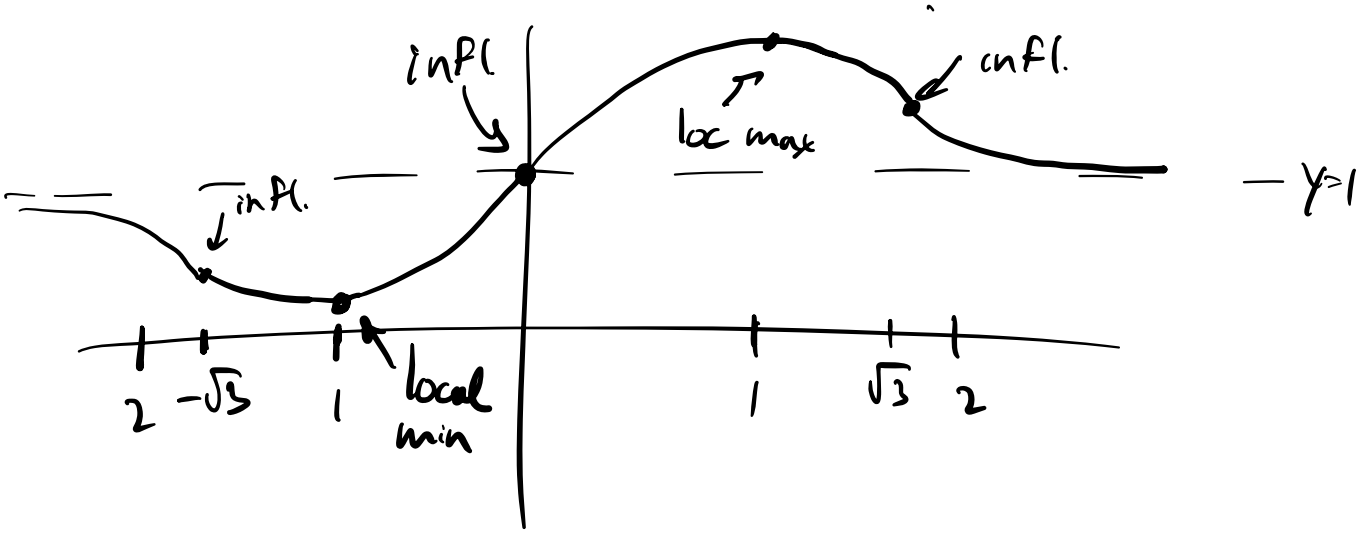
$$\lim_{x\to\infty} \frac{x}{x^2+1} + 1 = 1 + \lim_{x\to\infty} \frac{x \cdot \frac{1}{x^2}}{x^2+1 \cdot \frac{1}{x^2}} = 1 + \lim_{x\to\infty} \frac{1/x}{1+1/x^2}$$

$$= 1 + \frac{0}{1+0} = 1$$

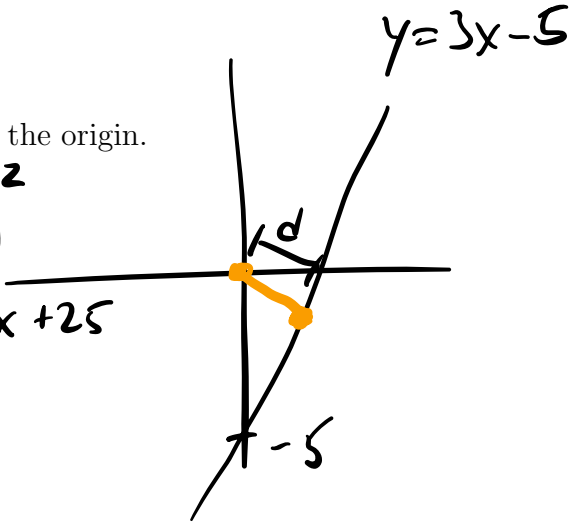
$$\lim_{x\to-\infty} \frac{x}{x^2+1} + 1 = 1 \text{ for same reason}$$

(remember: $\lim_{x\to\infty} 1/x^r = \lim_{x\to-\infty} 1/x^r = 0$ for $r>0$)

(e) Sketch the graph of $y = f(x)$, showing all the features above. Make sure to label the local maxima, local minima, and inflection points, and draw the asymptotes.



10. Find the point on the line $y = 3x - 5$ that is closest to the origin.



$d(x) = (\text{dist. from } (x, 3x-5) \text{ to } (0,0))^2$

$= x^2 + (3x-5)^2 = x^2 + 9x^2 - 30x + 25$

$= 10x^2 - 30x + 25$

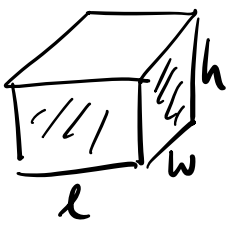
Minimize $d(x)$:

$d'(x) = 20x - 30 = 0 \iff 20x = 30 \iff x = 3/2$

This is a min. b/c $d'(x) < 0$ for $x < 3/2$, $d'(x) > 0$ for $x > 3/2$.

So $(3/2, 3 \cdot 3/2 - 5) = (3/2, -1/2)$ is closest to $(0,0)$.

11. If 15,000cm^2 of material is available to make a box with an open top and a square base, what is the largest possible volume of a box that can be made? What are the dimensions of the box with the largest volume? (You may leave your answers unsimplified.)



Square base $\rightsquigarrow \ell = w$

Surf. area $= \ell^2 + 4\ell h = 15{,}000$

So $h = \dfrac{15{,}000 - \ell^2}{4\ell}$

$V = \ell w h = \ell^2 \left(\dfrac{15000 - \ell^2}{4\ell}\right) = \dfrac{15000}{4}\ell - \dfrac{1}{4}\ell^3$

$\dfrac{dV}{d\ell} = \dfrac{15000}{4} - \dfrac{3}{4}\ell^2 = 0 \iff \dfrac{15000}{4} = \dfrac{3}{4}\ell^2 \iff \ell^2 = \dfrac{15000}{3} = 5{,}000$

$\iff \ell = \sqrt{5000}$

$V'' = -\dfrac{3}{2}\ell$ so $V'' < 0$ @ $\ell = \sqrt{5000}$ (note: $\ell > 0$ so $\ell \neq -\sqrt{5000}$)

$\rightsquigarrow \ell = \sqrt{\tfrac{1}{2} \cdot 10000} = \dfrac{100}{\sqrt{2}} = \dfrac{100\sqrt{2}}{2} = 50\sqrt{2}$ is a max

$V(\sqrt{5000}) = 5000 \cdot \dfrac{15{,}000 - 5000}{4\sqrt{5000}} = 5{,}000 \cdot \dfrac{10{,}000}{4} \cdot \dfrac{1}{50\sqrt{2}}$

$= \dfrac{1{,}000{,}000}{4} \cdot \dfrac{\sqrt{2}}{2} = 125{,}000\sqrt{2}\ \text{cm}^3$

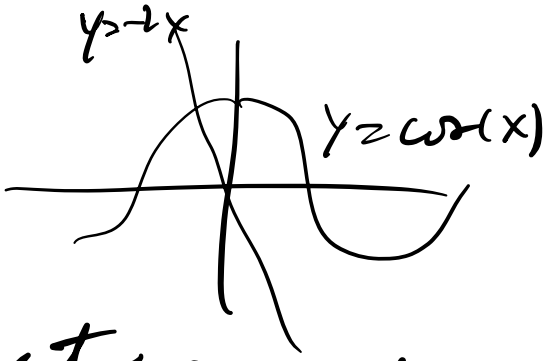
Bonus: Show that there are at most two solutions to the equation $\sin(x) = x^2$.

Say there were 3 sol'ns $x_1 < x_2 < x_3$.
Then Rolle's Thm says $f(x) = \sin(x) - x^2$
has $f'(x) = 0$ once in (x_1, x_2), and
once in (x_2, x_3) (at least).

But $f'(x) = -\cos(x) - 2x$

So $f'(x) = 0 \iff \cos(x) = -2x$

graph:



These graphs appear to cross only once.
If they crossed twice, Rolle's Thm
would say $g(x) = -\cos(x) - 2x$ has
$g'(x) = 0$ somewhere in between,
but $g'(x) = \sin(x) - 2$ is always < 0.